

- Q1) b) Data and a pointer to the next node
Q2) c) Null.
Q3) b) Self-referential structure
Q4) b) head == Null
Q5) b) newnode → next = head; head = newnode.
Q6) b) temp = head; head = head → next; free(temp);
Q7) c) Traversing from tail to head.
Q8) b) n
Q9) c) Copy data of P → next into P, then delete P → next.
Q10) b) while(P != Null).

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- Q11) b) P → next == Null.
Q12) c) Two pointers advancing at different rates.
Q13) c) Three
Q14) b) Exactly one node
Q15) b) P = (struct node*) malloc (Size of (struct node));
Q16) b) Avoid special case handling for insertion and deletion at head.
Q17) c) The first node
Q18) b) It does not support random (indexed) access.
Q19) b) Loss of access to all nodes, causing a memory leak.
Q20) b) A slow pointer and a fast pointer
Q21) c) The last node
Q22) b) Nodes may be scattered anywhere in the memory.

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Q23) struct node {
 int data;
 struct node* next;
};

Q24) ~~Ques~~

10 -> 20 -> 30 -> 40 -> Null

output:

30 50 70

10 + 20 = 30

20 + 30 = 50

30 + 40 = 70

Q25) void insertEnd(struct node **head, int x) {
 struct node *t = (struct node *) malloc
 (sizeof(struct node));
 t->data = x;
 t->next = Null;
 if (*head == Null) {
 *head = t;
 return;
 }
 struct node *p = *head;
 while (p->next != Null)
 p = p->next;
 p->next = t;
}

Q26) struct node* reverse(struct node* head) {
 struct node* prev = Null;
 struct node* current = head;
 struct node* next = Null;
 while (current != Null)
 {
 next = current->next;
 current->next = prev;
 prev = current;
 current = next;
 }
 return prev;
}

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28.) $5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{Null}$

(i) $7 \rightarrow 5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{Null}$

(ii) $7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow \text{Null}$

(iii) $7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow \text{Null}$